Explain Transimission impairment. Describe the block diagram of communication system.

In a communication system, the signal that is sent from the transmitter is rarely the same as the signal that arrives at the receiver. This degradation of signal quality is known as Transmission Impairment.

Transmission Impairment

There are three primary types of impairment that occur as a signal travels through a medium:

A. Attenuation

Attenuation means the loss of energy. When a signal travels through a medium (like a copper wire or air), it faces resistance, causing it to lose strength as distance increases.

- Solution: To compensate for this loss, amplifiers (for analog signals) or repeaters (for digital signals) are used to boost the signal.

B. Distortion

Distortion occurs when the shape or form of the signal changes. This is common in composite signals made of different frequencies. Since different frequencies travel at different speeds through a medium, they arrive at the destination at different times, causing the signal to change its original shape.

C. Noise

Noise refers to unwanted random energy that gets added to the signal from external sources.

- ✓ Thermal Noise: Random motion of electrons in a wire.
- ✓ Induced Noise: From external sources like motors or appliances.
- ✓ Crosstalk: Interference from one wire to another nearby wire.
- ✓ Impulse Noise: Sudden spikes from power lines or lightning.

Block Diagram of a Communication System

A basic communication system consists of several functional blocks that work together to transfer information from a source to a destination.

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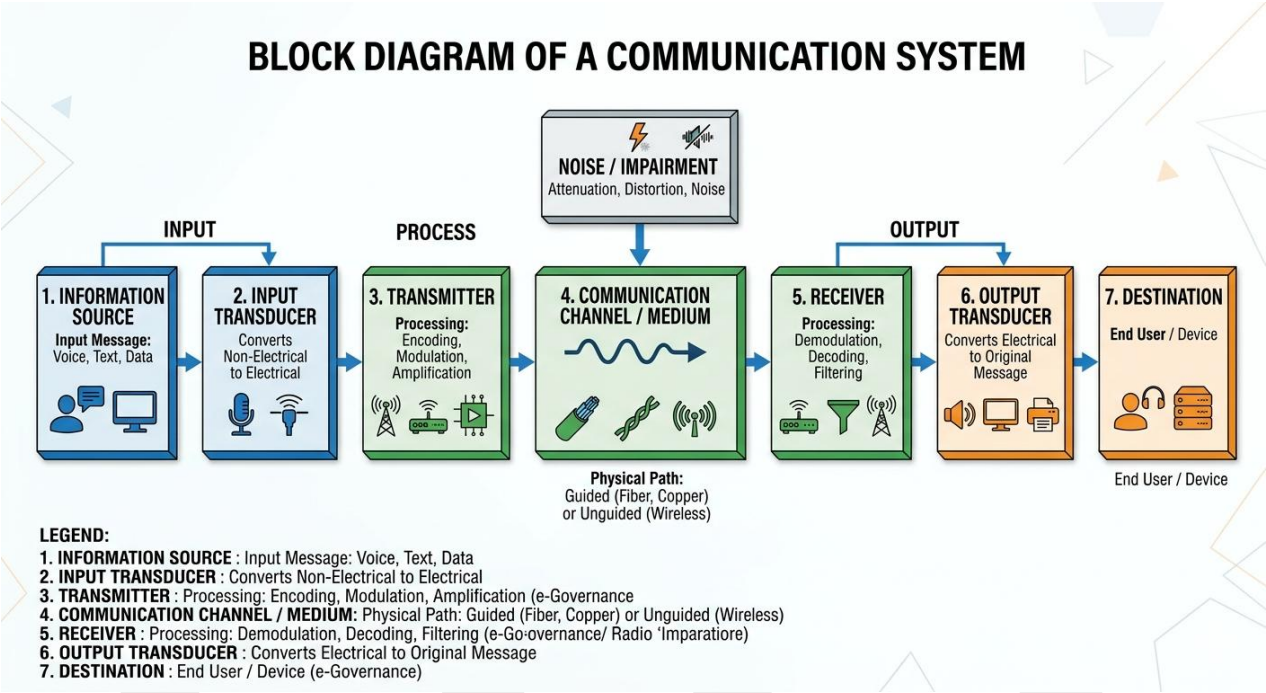

The Key Components: Information Source: * This is where the message is generated (e.g., human voice, a text file, or an image).

- ✧ **Input Transducer:** * It converts the non-electrical message (like sound) into an electrical signal suitable for processing.
- ✧ **Transmitter:** * The transmitter processes the electrical signal. Key functions include Modulation, Encoding, and Amplification to make the signal strong enough to travel long distances.
- ✧ **Communication Channel (Medium):** * The physical path between the transmitter and receiver. It can be Guided (cables, fiber optics) or Unguided (wireless/air). This is where Noise and Impairment occur.
- ✧ **Receiver:** * The receiver's job is to extract the original signal from the carrier. It performs Demodulation, Decoding, and Filtering to remove noise.
- ✧ **Output Transducer:** * **I**t converts the electrical signal back into its original form (e.g., a speaker converting electrical signals back into sound) so the user can understand it.
- ✧ **Destination:** * The final stage where the message is received by the intended person or machine.

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Differentiate between MAN and WAN. Explain OSI referent model.

1. Difference Between MAN and WAN

MAN (Metropolitan Area Network) and **WAN (Wide Area Network)** are categorized based on their geographical span and data transfer speeds.

Feature	MAN (Metropolitan Area Network)	WAN (Wide Area Network)
Area Covered	Covers a city or a large campus.	Covers a country, continent, or the whole world.
Ownership	Usually owned by a single organization or a local ISP.	Owned by multiple organizations (Public/Private).
Speed	High speed (Faster than WAN).	Lower speed (due to long distance).
Propagation Delay	Moderate.	High.
Cost	Moderately expensive.	Very high setup and maintenance cost.
Example	Cable TV network in a city, City-wide Wi-Fi.	The Internet, 4G/5G mobile networks.

2. OSI Reference Model

The **OSI (Open Systems Interconnection)** model is a conceptual framework developed by the ISO that describes how data from an application on one computer moves through a network medium to an application on another computer. It consists of 7 layer.

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The Upper Layers (Host Layers)

These layers deal with user interface, data formatting, and session management.

7. Application Layer: The layer closest to the user. It provides network services to applications (e.g., HTTP for web browsing, SMTP for email).

6. Presentation Layer: Acts as a translator. It handles data encryption, compression, and formatting (e.g., converting JPG or MP3 into a format the application layer understands).

5. Session Layer: Manages the "session" or dialogue between two computers. It establishes, maintains, and terminates connections.

The Lower Layers (Network Layers)

These layers deal with the actual movement of data across the physical medium.

4. Transport Layer: Responsible for end-to-end communication. It breaks data into **Segments** and ensures reliable delivery using protocols like TCP (Transmission Control Protocol).

3. Network Layer: Determines the best physical path for the data. It deals with **Packets** and uses logical addressing (IP Addresses). Routers operate at this layer.

2. Data Link Layer: Provides error-free transfer of data frames over the physical link. It deals with **Frames** and physical addressing (MAC Addresses). Switches operate here.

1. Physical Layer: The lowest layer, dealing with the actual physical connection (cables, radio waves). It transmits raw **Bits** (0s and 1s) over the medium.

Mnemonics to Remember the OSI Layers:

To remember the layers from top (7) to bottom (1):

> **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing
>
> *(Application, Presentation, Session, Transport, Network, Data Link, Physical)*

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What is a public switched telephone network? Explain fiber optic cable with its applications.

In the world of telecommunications, the **PSTN** and **Fiber Optic Cables** represent the evolution from traditional voice calls to high-speed digital data transmission.

Public Switched Telephone Network (PSTN)

The **PSTN** is the aggregate of the world's circuit-switched telephone networks that are operated by national, regional, or local telephony operators. It is the infrastructure that allows any phone in the world to connect to any other phone.

How it Works:

- **Circuit Switching:** When you make a call, a dedicated physical path (circuit) is established between the two phones for the duration of the call.
- **Nodes and Switches:** The network consists of telephone lines, fiber optic cables, microwave transmission links, cellular networks, and switching centers.
- **Standardization:** It follows international standards (like the ITU-T) to ensure that a call made in Nepal can reach a recipient in the USA seamlessly.
- **Evolution:** While it started with analog copper wires and manual switchboards, modern PSTN is almost entirely digital in its core, using cellular and fiber technology.

Fiber Optic Cable

A **Fiber Optic Cable** is a high-speed data transmission medium that contains strands of glass fibers inside an insulated casing. Unlike copper wires that transmit data using electrical pulses, fiber optics transmit data as **pulses of light**.

How it Works (Total Internal Reflection):

Data is converted into light by a laser or LED. This light travels through the **Core** (the center glass strand) by bouncing off the **Cladding** (the outer layer). Because the cladding has a lower refractive index, the light stays trapped inside the core, traveling at incredible speeds with very little loss of signal.

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Applications of Fiber Optic Cables

- **Internet Connectivity:** It is the backbone of the Global Internet. Submarine fiber cables connect continents, and "Fiber to the Home" (FTTH) provides high-speed internet to residents.
- **Telecommunications:** Most modern telephone systems (PSTN) and cellular backhaul (connecting towers to the core network) use fiber for clear, lag-free voice calls.
- **Medical Imaging:** Used in **Endoscopy** and **Laparoscopy**, where thin fiber bundles carry light into the body and transmit images back to a screen for surgery.
- **Military and Aerospace:** Used in fighter jets and naval ships for secure, high-speed communication that cannot be jammed by electronic warfare.
- **Industrial Sensors:** Used to monitor temperature and pressure in harsh environments (like oil rigs) where electrical sensors might cause a fire or fail due to heat.
- **Cable TV (CATV):** Used by cable providers to deliver high-definition video signals with much higher clarity than old coaxial cables.

What is forming? Explain the issues of data link layer.

What is Framing?

In the Data Link Layer, **Framing** is the process of breaking down the stream of bits received from the Network Layer into manageable, discrete blocks called **Frames**.

Since the Physical Layer just sends a continuous stream of 0s and 1s, the receiver needs a way to know where one message ends and the next one begins. Framing adds specific header and trailer information to each block of data to achieve this.

Common Methods of Framing:

- **Character Count:** The header specifies the total number of characters in the frame. (Risk: If the count number gets corrupted, the entire synchronization is lost).
- **Flag bits / Bit Stuffing:** Special bit patterns (like 01111110) are used as "flags" to mark the start and end of a frame.

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- **Physical Layer Coding Violations:** Used in networks like Ethernet, where specific electrical signals (that don't represent 0 or 1) are used to mark boundaries.

Issues of the Data Link Layer

The Data Link Layer is responsible for moving data between two nodes on the same network. To do this reliably, it must handle several technical "issues" or challenges:

1. Framing & Synchronization

As mentioned above, the primary issue is defining the boundaries of data. If the start or end of a frame is not recognized correctly, the receiver will interpret the data as garbage.

2. Error Control

Physical transmission is never 100% perfect. Signals can be distorted by noise.

- **The Issue:** How does the receiver know if a bit changed from 0 to 1 during transit?
- **The Solution:** The layer uses techniques like CRC (Cyclic Redundancy Check) or **C**hecksums in the trailer to detect errors. If an error is found, it asks the sender to retransmit the frame.

3. Flow Control

Imagine a high-speed server sending data to an old, slow computer.

- **The Issue:** If the sender transmits frames faster than the receiver can process them, the receiver's buffer will overflow, and data will be lost.
- **The Solution:** Mechanisms like **Stop-and-Wait** or **Sliding Window** protocols are used to balance the speed between sender and receiver.

4. Access Control (Medium Access Control - MAC)

On a shared medium (like a Wi-Fi network or a single Ethernet cable), multiple devices might try to send data at the exact same time.

- **The Issue:** If two devices "talk" at once, a **Collision** occurs, and both messages are destroyed.
- **The Solution:** Protocols like **CSMA/CD** (for wired) or **CSMA/CA** (for wireless) determine whose turn it is to speak.

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5. Physical Addressing

Since a network may have many devices, the Data Link Layer must ensure the frame reaches the correct hardware.

- **The Issue:** Differentiating between devices on the same local link.
- **The Solution:** It uses **MAC Addresses** (Unique 48-bit hardware addresses) to identify the source and destination.

What is Ethernet? Explain ALOHA and VLAN

Ethernet is the most widely used technology for connecting computers in a **Local Area Network (LAN)**. It defines the wiring and signaling standards for the Physical Layer and the framing protocols for the Data Link Layer of the OSI model.

- ✓ Standard: It is officially standardized as IEEE 802.3.
- ✓ Medium: Originally used coaxial cables, but modern Ethernet uses Twisted Pair (RJ-45) or Fiber Optic cables.
- ✓ Access Method: It traditionally uses CSMA/CD (Carrier Sense Multiple Access with Collision Detection) to manage how devices share the cable.
- ✓ Speed: Evolves from Standard Ethernet (10 Mbps) to Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps), and beyond.

2. ALOHA

ALOHA is one of the earliest "Random Access" protocols, developed at the University of Hawaii. It was designed to solve the problem of how multiple stations can use a shared communication channel without a central controller.

There are two main versions:

A. Pure ALOHA

- ✓ Rule: A station transmits a frame whenever it has data to send.
- ✓ Collision: If two stations transmit at the same time, a collision occurs, and the frames are destroyed. The stations then wait for a random amount of time and try again.
- ✓ Efficiency: Very low (maximum throughput is only 18.4%) because collisions can happen at any time.

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B. Slotted ALOHA

- ✓ **Rule:** Time is divided into discrete "slots." A station can only start transmitting at the **beginning** of a time slot.
- ✓ **Efficiency:** By forcing stations to synchronize, the "vulnerable period" is reduced by half. This doubles the maximum throughput to **36.8%**.

3. VLAN (Virtual Local Area Network)

A VLAN is a logical group of workstations, servers, and network devices that appear to be on the same LAN despite their geographical distribution. It allows a network administrator to segment a network without needing to run new cables or change physical switches.

Why do we use VLANs?

- ✓ Segmentation: You can separate the "Accounting Department" from the "Student Wi-Fi" on the same physical switch so they cannot see each other's traffic.
- ✓ Security: Sensitive data can be isolated into a private VLAN.
- ✓ Reduced Broadcast Traffic: In a large network, "broadcast" messages can slow everything down. VLANs keep these messages within a smaller logical group.
- ✓ Flexibility: If an employee moves to a different office in the same building, the administrator can simply change their port assignment in the software rather than rewiring the office.

Calculate network address, usable host address number of subnets and boardcast address for a network of 192.168.1.20/26

To calculate the network parameters for the address **$192.168.1.20/26$**, we first need to break down the CIDR notation ($/26$) and the IP address.

1. Breakdown of CIDR (/26)

The prefix **$/26$** means that the first 26 bits of the 32-bit IPv4 address are the **Network Bits**, and the remaining 6 bits ($32 - 26 = 6$) are the **Host Bits**.

- ✓ Subnet Mask (Binary): $11111111.11111111.11111111.11000000$
- ✓ Subnet Mask (Decimal): $255.255.255.192$

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2. Calculations

A. Network Address

The network address is found by setting all **Host Bits** to 0. Since 26 bits are for the network, we only look at the last octet ($192.168.1.\mathbf{20}$).

In a $/26$ subnet, the block size is $2^{(32-26)} = 2^6 = 64$.

The subnets in the last octet will be multiples of 64: $0, 64, 128, 192$.

Since $.20$ falls between 0 and 63, the start of the range is $.0$.

- **Network Address: $192.168.1.0$**

B. Broadcast Address

The broadcast address is the last address in the subnet (where all Host Bits are 1). It is one address before the next subnet ($64 - 1 = 63$).

Broadcast Address: $192.168.1.63$

C. Usable Host Addresses

Usable hosts are those between the Network Address and the Broadcast Address.

- ✓ First Usable Host: $192.168.1.1$
- ✓ Last Usable Host: $192.168.1.62$
- ✓ Total Usable Host Count: $2^6 - 2 = \mathbf{62}$

D. Number of Subnets

Assuming the network started as a standard **Class C** network ($/24$), we calculate the subnets based on the bits "borrowed" from the host portion.

- ✓ **Bits Borrowed:** $26 - 24 = 2$ bits.
- ✓ **Number of Subnets:** $2^2 = \mathbf{4}$ subnets.

Explain quality of service. Differentiate between TCP and UDP.

1. Quality of Service (QoS): **Quality of Service (QoS)** is a set of technologies and techniques used to manage network resources by prioritizing specific types of data traffic. Since network bandwidth is finite, QoS ensures that high-priority applications get the resources they need to perform well.

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Why do we need QoS?

Without QoS, data is handled on a "Best Effort" basis, meaning all packets are treated equally. This causes problems for real-time services like:

- ✓ **Latency (Delay):** The time it takes for a packet to travel from source to destination. High latency causes "lag" in video calls.
- ✓ **Jitter:** Variation in the delay of received packets. This causes audio to sound "choppy."
- ✓ **Packet Loss:** When packets are dropped due to network congestion.

2. Difference Between TCP and UDP

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Type	**Connection-Oriented:** Establishes a link before sending data (Handshake).	**Connectionless:** Sends data without checking if the receiver is ready.
Reliability	**High:** Guaranteed delivery. It retransmits lost packets.	**Low:** No guarantee. If a packet is lost, it stays lost.
Order	Delivers data in the **exact sequence** it was sent.	Data may arrive out of order.
Speed	**Slower** due to the overhead of error checking and acknowledgments.	**Faster** because it has no error recovery or flow control.
Data Unit	Segment.	Datagram.
Usage	Web browsing (HTTP), Email (SMTP), File transfer (FTP).	Video streaming, Online gaming, VoIP, DNS.

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What is PoP3? Differentiate between HTTP and HTTPS.

POP3 (Post Office Protocol version 3) is a standard "client-server" protocol used to retrieve emails from a remote server to a local email client (like Outlook, Thunderbird, or a mobile mail app).

It is called a "Post Office" protocol because it works much like a physical post office:

- ✓ Storage: The server receives and stores your emails.
- ✓ Collection: You connect to the server and download your messages to your device.
- ✓ Deletion: Once downloaded, the emails are typically deleted from the server and stored only on your local device.

Key Characteristics of POP3:

- Offline Access: Since emails are downloaded to your computer, you can read them even without an internet connection.
- Server Space: It saves space on the mail server because messages are removed after being downloaded.
- Single Device Focus: Because it deletes messages from the server, it is difficult to sync emails across multiple devices (e.g., if you download an email on your PC, you won't see it on your phone later).

2. Difference Between HTTP and HTTPS

Feature	**HTTP**	**HTTPS**
Full Form	Hypertext Transfer Protocol	Hypertext Transfer Protocol **Secure**
Security	**Unsecured.** Data is sent in "Plain Text."	**Secured.** Data is **Encrypted** using SSL/TLS.
Port Number	Uses Port **80**.	Uses Port **443**.
Privacy	Hackers can "sniff" or read your data (passwords, credit cards) easily.	Even if intercepted, the data is unreadable to hackers.
Authentication	No identity verification of the website.	Uses **Digital Certificates** to prove the website is genuine.
Browser Indicator	Shows "Not Secure" in the address bar.	Shows a **Padlock icon** and starts with https://.

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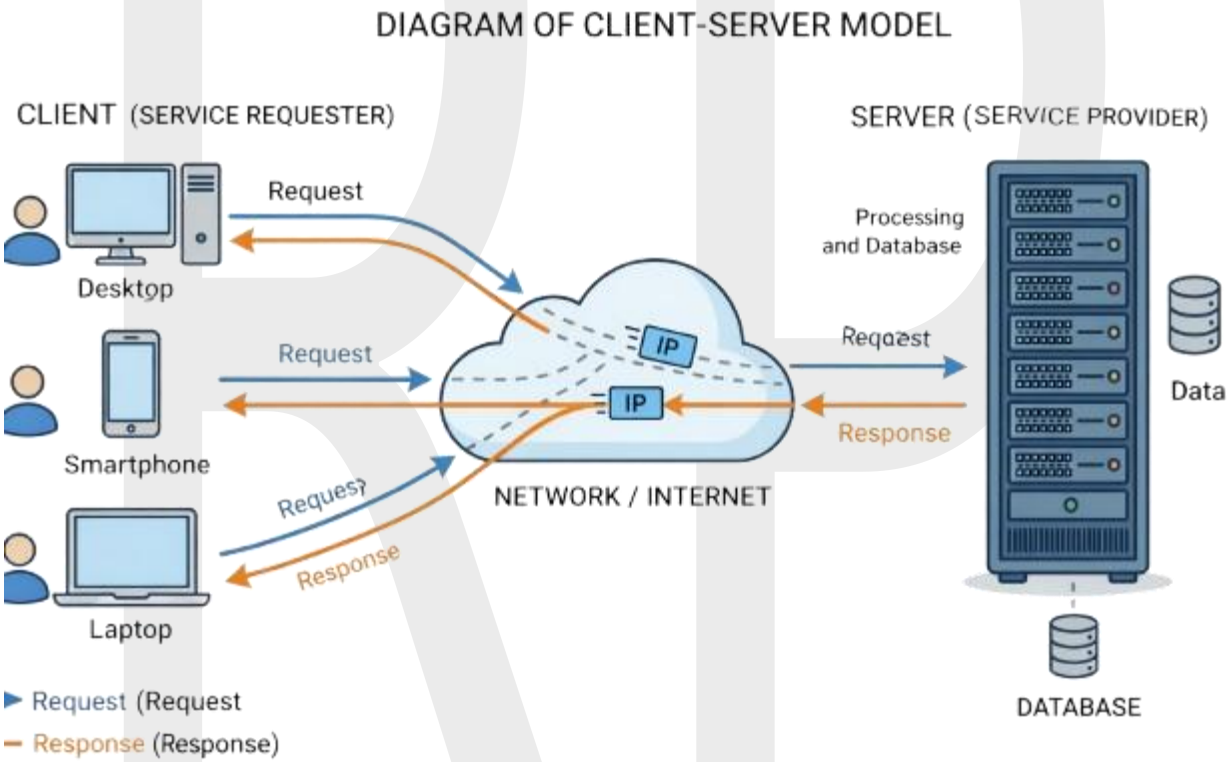
What is Router? Describe the client-server model with figure

A Router is a networking device that forwards data packets between computer networks. It operates at the Network Layer (Layer 3) of the OSI model.

Think of a router as a "Traffic Controller" or a "Post Office" for the internet. It looks at the destination IP address of a data packet and uses a Routing Table to determine the best and fastest path for that packet to reach its destination.

Client-Server Model

The Client-Server Model is a distributed application structure that partitions tasks or workloads between the providers of a resource or service (called Servers) and service requesters (called Clients).



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What is IP Address? List the types of IP Address.

An IP Address (Internet Protocol Address) is a unique numerical label assigned to every device (computer, smartphone, printer, router) connected to a computer network that uses the Internet Protocol for communication.

Think of it like a postal address for your digital device. Just as a physical house needs an address so the mailman knows where to deliver a letter, a computer needs an IP address so the network knows where to send data packets.

Types of IP Addresses

IP addresses are categorized in several ways based on their version, how they are assigned, and their visibility.

1. Based on Version

- ✓ **IPv4 (Internet Protocol version 4):** The most common version. It uses a 32-bit address format (e.g., 192.168.1.1). It allows for about 4.3 billion unique addresses.
- ✓ **IPv6 (Internet Protocol version 6):** Developed because the world ran out of IPv4 addresses. It uses a 128-bit hexadecimal format (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334). It provides an almost infinite number of addresses.

2. Based on Visibility (Scope)

- ✓ **Public IP Address:** The address assigned to your network router by your ISP. It is visible to the entire internet and must be unique globally.
- ✓ **Private IP Address:** Used within a local network (like your home or office Wi-Fi). Devices inside the network use these to talk to each other but are hidden from the outside world.

3. Based on Assignment Method

- ✓ **Static IP Address:** An address that is manually assigned and **never changes**. These are typically used for servers, printers, or web hosting.
- ✓ **Dynamic IP Address:** An address that is automatically assigned by a **DHCP server** and changes periodically (e.g., when you restart your router). Most home users have dynamic IPs.

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4. Special Types

- ✓ **Loopback Address (127.0.0.1):** Used by a computer to talk to itself (mostly for testing software).
- ✓ **Broadcast Address:** Used to send data to all devices on a specific subnet simultaneously.

Write short notes

DNS

Domain Name System (DNS): DNS is often called the **"Phonebook of the Internet."** It is a hierarchical and decentralized naming system used to translate human-readable domain names (like google.com) into machine-readable **IP addresses** (like 142.251.46.238).

Since computers and routers communicate using numbers (IPs), but humans find it easier to remember words, DNS acts as the bridge between the two.

How DNS Works (The Resolution Process)

When you type a URL into your browser, the following steps occur:

- ✓ DNS Query: Your computer checks its local cache. If not found, it sends a request to a Recursive Resolver (usually provided by your ISP).
- ✓ Root Nameserver: The resolver asks the Root Server, "Where is the .com registry?"
- ✓ TLD Nameserver: The Root points to the Top-Level Domain (TLD) server (e.g., for .com, .org, .np).
- ✓ Authoritative Nameserver: The TLD server points to the specific server that holds the actual IP record for that domain.
- ✓ IP Retrieval: The Authoritative server provides the IP address back to the resolver, which then gives it to your browser to load the website.

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HDLC

High-Level Data Link Control (HDLC)

HDLC is a bit-oriented, synchronous data link layer protocol developed by the International Organization for Standardization (ISO). It is used to provide reliable delivery of data over a communication link, ensuring that the data received is exactly what was sent.

It is widely used in point-to-point and point-to-multipoint configurations and serves as the basis for many other protocols like PPP (Point-to-Point Protocol).

1. HDLC Frame Structure

HDLC uses a specific frame format to manage data transmission. Every frame starts and ends with a unique "Flag" sequence.

- ✓ **Flag (8 bits):** The sequence 01111110 marks the beginning and end of a frame.
- ✓ **Address (8 or more bits):** Identifies the secondary station that is sending or receiving the frame. In point-to-point links, it is often set to a broadcast address.
- ✓ **Control (8 or 16 bits):** This is the "brain" of the frame. It manages flow control and error control (sequence numbers, ACKs).
- ✓ **Information (Variable):** Contains the actual user data from the network layer.
- ✓ **FCS (Frame Check Sequence - 16 or 32 bits):** Uses **CRC (Cyclic Redundancy Check)** to detect if bits were corrupted during transmission.

Data Transfer Modes

HDLC supports three modes of operation:

- ✓ **Normal Response Mode (NRM):** Used in unbalanced configurations (one primary station, many secondaries). Secondaries can only transmit when given permission by the primary.
- ✓ **Asynchronous Response Mode (ARM):** Similar to NRM, but a secondary can initiate transmission without explicit permission.
- ✓ **Asynchronous Balanced Mode (ABM):** The most common mode for point-to-point links. Both stations act as equals (combined stations) and can initiate transmission at any time.

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Guided Media

It looks like you're asking about Guided Media (also known as Wired or Bound transmission media). In data communication, guided media are physical paths through which signals are directed using a solid medium.

Here are the three main types used in networking:

1. Twisted Pair Cable

This is the most common type of guided media, consisting of two insulated copper wires twisted together. The twisting helps reduce Crosstalk and electromagnetic interference.

- Unshielded Twisted Pair (UTP): Common in Ethernet networks (like the Cat5e or Cat6 cables you likely use).
- Shielded Twisted Pair (STP): Has a metal foil covering to provide extra protection against external noise.
- Application: Local Area Networks (LANs) and traditional telephone lines.

2. Coaxial Cable

Coaxial cable has a central copper conductor surrounded by an insulating layer, a metallic shield (braid), and an outer plastic jacket.

- **Performance:** It carries signals of higher frequency ranges than twisted pair and is less susceptible to interference.
- **Application:** Cable TV (CATV) networks, traditional Ethernet (Thicknet/Thinnet), and high-speed cable internet.

3. Fiber Optic Cable

Fiber optics use pulses of light to transmit data through thin strands of glass or plastic. It works on the principle of Total Internal Reflection.

Advantages:

- Extremely high bandwidth and speed.
- Immune to Electromagnetic Interference (EMI).
- Supports very long distances without signal loss (attenuation).

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Application: Internet backbones, undersea cables connecting continents, and high-speed FTTH (Fiber to the Home) connections.

Analog Data Tranmission

In data communication, **Analog Data Transmission** refers to the transfer of information using continuous signals. Unlike digital signals, which use discrete 0s and 1s, analog signals vary continuously in terms of amplitude and frequency over time.

1. Nature of Analog Signals

Analog data is typically represented by a **Sine Wave**. To describe an analog signal, we look at three main characteristics:

- ✓ **Amplitude:** The height of the wave, representing the signal's strength or intensity.
- ✓ **Frequency:** The number of cycles the wave completes in one second (measured in Hertz, Hz).
- ✓ **Phase:** The position of the waveform relative to time zero.

2. How Analog Transmission Works

In this method, the "data" (like your voice) is converted into an electrical signal that mimics the original physical medium.

- ✓ **Continuous Flow:** The signal has an infinite number of values within a range.
- ✓ **Media:** It is commonly transmitted over guided media like **twisted-pair copper wires** (telephone lines) or unguided media like **radio waves**.
- ✓ **Amplification:** Because signals lose strength over distance (**Attenuation**), analog systems use **Amplifiers** to boost the entire signal.

Note: A major drawback is that amplifiers also boost any "noise" that has joined the signal, which can lead to distortion.

3. Modulation (Converting Data for Transmission)

Since pure analog data (like a human voice) cannot travel very far on its own, it is often "carried" by a higher-frequency signal. This process is called **Modulation**.

- ✓ **Amplitude Modulation (AM):** The amplitude of the carrier wave is varied to match the data.

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- ✓ **Frequency Modulation (FM):** The frequency of the carrier wave is changed according to the data.
- ✓ **Phase Modulation (PM):** The phase of the carrier wave is shifted to represent the data.

Properties of secure communication

To ensure that information is exchanged safely over a network, a system must satisfy specific security requirements. In the field of cyber security and data communication, these are often referred to as the CIA Triad, with a few additional pillars added for modern networks.

Here are the five essential properties of secure communication:

1. Confidentiality

Confidentiality ensures that only the intended recipient can read the message. Even if a hacker intercepts the data, they should not be able to understand its contents.

- ✓ How it's achieved: Primarily through Encryption (converting plain text into cipher text using algorithms like AES or RSA).
- ✓ Example: When you send a private message, only you and your friend have the "keys" to read it.

2. Integrity

Integrity guarantees that the message has not been altered, deleted, or tampered with during transmission. The receiver must be sure that the data they see is exactly what the sender sent.

- ✓ How it's achieved: Using Hash functions (like SHA-256) or Digital Signatures. If even one bit of the data changes, the "hash" value will change, alerting the receiver.
- ✓ Example: Ensuring that a bank transfer for Rs. 1,000 doesn't get changed to Rs. 10,000 by a middleman.

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3. Authentication

Authentication is the process of verifying the identity of the parties involved in the communication. The receiver needs to be sure the message really came from the claimed sender, and the sender needs to know they are talking to the correct receiver.

- ✓ How it's achieved: Through Passwords, Digital Certificates, or Biometrics.
- ✓ Example: Logging into your e-Governance portal requires a username and password to prove it's really you.

4. Availability

Availability ensures that the communication system and the data are accessible to authorized users whenever they are needed. A secure system is useless if it is constantly crashing or blocked by attacks.

- ✓ **How it's achieved:** Using **Firewalls**, **DDoS protection**, and **Redundant systems** (backup servers).
- ✓ **Example:** Ensuring that a government tax filing website stays online even during high-traffic periods near the deadline.

5. Non-Repudiation

Non-repudiation prevents a sender from later denying that they sent a specific message. Similarly, a receiver cannot deny having received it. It provides legal "proof" of the transaction.

- ✓ **How it's achieved:** Using **Digital Signatures** and **Public Key Infrastructure (PKI)**. Since only the sender has their private key, only they could have signed the message.
- ✓ **Example:** Once you digitally sign a contract or a land registry document, you cannot claim later that "I didn't send that."

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